Project Planning Phase
Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	15 February 2025
Team ID	LTVIP2025TMID36086
Project Name	Enchanted Wings: Marvels of Butterfly Species
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	2	High	
Sprint-1		USN-2	As a user, I will receive confirmation email once I have registered for the application	1	High	
Sprint-2		USN-3	As a user, I can register for the application through Facebook	2	Low	
Sprint-1		USN-4	As a user, I can register for the application through Gmail	2	Medium	
Sprint-1	Login	USN-5	As a user, I can log into the application by entering email & password	1	High	
	Dashboard					

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

https://www.visual-paradigm.com/scrum/scrum-burndown-chart/
https://www.atlassian.com/agile/tutorials/burndown-charts

Reference:
https://www.atlassian.com/agile/project-management
https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software
https://www.atlassian.com/agile/tutorials/epics
https://www.atlassian.com/agile/tutorials/sprints
https://www.atlassian.com/agile/project-management/estimation
https://www.atlassian.com/agile/tutorials/burndown-charts

✅ Product Backlog, Sprint Schedule & Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint -1	Data Collection	USN-1	As a researcher, I can collect butterfly images from field sources and datasets.	2	High	Team Member A

Sprint -1	Data Collection	USN-2	As a user, I can load the images into project folders based on species.	1	High	Team Member B
Sprint -1	Data Preparation	USN-3	As a data analyst, I can handle missing/corrupted files before training.	3	High	Team Member A
Sprint -1	Data Preparation	USN-4	As a data engineer, I can create proper folder/class structure for training.	3	Medium	Team Member B
Sprint -1	Data Preparation	USN-5	As a user, I can remove duplicates/inconsistencies in data for better accuracy.	3	Medium	Team Member C
Sprint -2	Data Visualization	USN-6	As a user, I can visualize class distribution using bar, pie, and line charts.	2	Medium	Team Member B
Sprint -2	Dashboard Development	USN-7	As a user, I can use a dashboard to upload butterfly images and see predictions.	5	High	Team Member A
Sprint -2	Story Generation	USN-8	As a user, I can generate a final story/report of prediction results.	5	High	Team Member C

Sprint -2	Data Visualization	USN-9	As a user, I can plot prediction results on a map interface.	3	Low	Team Member B

📈 Project Tracker, Velocity & Burndown Chart (4 Marks)

📊 Sprint Tracking Table

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	12	6 Days	10 Feb 2025	15 Feb 2025	12	15 Feb 2025
Sprint-2	20	6 Days	17 Feb 2025	22 Feb 2025	20	22 Feb 2025

⚙️ Velocity Calculation

Velocity=Total Story Points CompletedNumber of Sprints=12+202=16 Story Points/Sprint\text{Velocity} = \frac{\text{Total Story Points Completed}}{\text{Number of Sprints}} = \frac{12 + 20}{2} = 16 \text{ Story Points/Sprint}Velocity=Number of SprintsTotal Story Points Completed=212+20=16 Story Points/Sprint

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📐 Velocity per Day (Estimation)

For a 6-day sprint:

$$\text{Average Velocity (AV)} = \frac{16\text{ SP}}{6\text{ days}} \approx 2.67\text{ SP/day}$$

📉 Burndown Chart Overview

A burndown chart will show story points remaining (y-axis) over time (x-axis), gradually approaching 0 by end of sprint.

For Sprint-2, you can sketch or generate this graph based on the 20 SP planned and daily completion (e.g., 3–4 SP/day).

📎 References

- [Atlassian Agile Project Management]
- [Scrum Tutorials - JIRA]
- [Burndown Charts Guide]